

SemanticCanvas Extension: Speech-to-Text (STT) Integration

1. Overview

This extension introduces Speech-to-Text (STT) capabilities to the SemanticCanvas environment. It allows users to dictate text directly into "Text Things" and provides a configurable pipeline for various transcription engines (Local, Remote, or Browser-native).

2. Configuration and Settings

2.1 STT Model Profiles

The "Models & API" settings section shall be expanded to include a dedicated tab for **STT Profiles**. Each profile defines a specific transcription provider and its parameters.

- **Provider Types:**
 - **Local:** Integration with local STT engines (e.g., OpenAI Whisper running via Ollama or a dedicated local FastAPI endpoint).
 - **Remote (API):** Support for cloud-based providers such as OpenRouter, OpenAI (Whisper API), or Deepgram.
 - **Browser-Native:** Utilizes the standard Web Speech API available in modern browsers (zero-cost, no server-side processing).
- **Profile Parameters:**
 - Profile Name: Unique identifier for the configuration.
 - Endpoint URL: (For Local/Remote) The API target. Defaults to <https://openrouter.ai/api/v1/audio/transcriptions> for OpenRouter profiles.
 - API Key: (For Remote) Secure credential storage.
 - Model ID: Specific model string. For OpenRouter, supported models include:
 - openai/whisper-large-v3
 - openai/whisper-large-v3-turbo
 - groq/whisper-large-v3
 - groq/whisper-large-v3-turbo
 - groq/distil-whisper-large-v3-en
 - Language: ISO code (e.g., en-US, fr-FR) or Auto-detect.
 - Temperature: Sampling temperature for probabilistic models (0.0 to 1.0).
 - Prompt: (Optional) Contextual text to guide the model's style or terminology.

2.2 Global Defaults

In the general AI settings, a **Default STT Profile** must be selectable. This profile is automatically

assigned to any new Canvas or Text Thing unless explicitly overridden.

3. User Interface (UI) Updates

3.1 Top Panel Controls

The Canvas top panel, which currently houses LLM and VLM selection dropdowns, shall be updated to include an **STT Selector**.

- **Component:** A searchable dropdown menu.
- **Function:** Allows the user to change the active STT configuration for the entire canvas session.
- **Visuals:** Prefixed with a Mic icon (Lucide-react: mic).

3.2 Text Thing (Node) Integration

The "Text" type node is enhanced with direct voice input capabilities.

- **Microphone Icon:** A small button located in the node's toolbar or footer.
- **Interactive States:**
 - **Idle:** Standard gray mic icon.
 - **Active/Listening:** Pulsing red icon or a glowing ring around the node to indicate the microphone is hot.
 - **Processing:** A loading spinner replaces the mic icon while the audio is being transcribed by the backend.
- **Input Logic:**
 - Single-click to start/stop (Toggle).
 - Transcription results are appended to the current text content or inserted at the cursor position if the node is in edit mode.

4. Technical Implementation

4.1 Backend Engine (FastAPI)

The backend must handle audio streaming or file-based uploads depending on the provider.

- **Audio Pre-processing:** If using Whisper, the backend should handle basic noise reduction and normalization.
- **Async Handling:** Transcription requests must be handled as asynchronous tasks to prevent blocking the UI.
- **OpenRouter / OpenAI Integration:**
 - Requests are sent as multipart/form-data.
 - Mandatory fields: file (Blob/File), model.
 - Optional fields: language, prompt, response_format (default: json), temperature.
- **Provider Adapters:**

- STTProviderInterface: Abstract class ensuring consistent output (Text + Confidence Score) regardless of the provider.

4.2 Browser Implementation

For the Browser-Native provider:

- Use window.SpeechRecognition (or webkitSpeechRecognition).
- Implement continuous listening logic if requested by the user.
- **Limitation Handling:** Provide UI feedback if the browser does not support the Web Speech API or if microphone permissions are denied.

4.3 Data Model Updates (SQLite)

- **Table stt_configs:**
 - id (PK)
 - name (String)
 - provider_type (Enum: LOCAL, REMOTE, BROWSER)
 - api_endpoint (String, nullable)
 - api_key (Encrypted String, nullable)
 - model_id (String)
 - language_code (String)
 - is_default (Boolean)

5. Security & Privacy

- **Permission Management:** Explicit browser permission prompts must be triggered only when the user clicks the microphone icon.
- **Data Privacy:** Local profiles must ensure audio data never leaves the local network. Remote profiles (OpenRouter, OpenAI) must comply with the general application RBAC and security protocols.
- **Sensitive Data:** Implement a "Mute" indicator and an easy-access global stop button in the top panel to kill all active STT sessions.